

Kyong Hwan Jin (진경환)

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EDUCATION

Ph.D., Bio and Brain Engineering, Korea Advanced Institute of Science and Technology (KAIST), South Korea, 2015 (Advisor: Jong Chul Ye; Thesis: *High-speed imaging system and interpolation algorithm for irregularly undersampled data*) | B.S., Bio and Brain Engineering, KAIST, South Korea, 2008

PROFESSIONAL EXPERIENCE

- **2023.09–present:** Associate Professor, School of Electrical Engineering, Korea University
- **2021.02–2023.08:** Assistant Professor, Dept. of Electrical Engineering and Computer Science (EECS), DGIST
- **2019.09–2021.02:** Staff Engineer, Samsung Research, Camera T/F – Global AI Center
- **2016.06–2019.08:** Postdoctoral Researcher, École polytechnique fédérale de Lausanne (EPFL), Biomedical Imaging Group (PI: Michael Unser)
- **2015.03–2016.05:** Postdoctoral Researcher, KAIST, Bioimaging Signal Processing Lab (PI: Jong Chul Ye)

RESEARCH INTERESTS

Machine learning, generative models (diffusion), signal/image processing, inverse problems, agentic AI/world model — with a focus on bridging the mathematical rigor of classical signal processing with modern deep-learning representations.

SELECTED RESEARCH CONTRIBUTIONS

- **Continuous-domain signal decoding for imaging:** developed JDEC, extending the block-wise discrete-cosine decoding at the core of JPEG to continuous cosine coefficients, removing blocking/ringing artifacts and enabling arbitrary-resolution reconstruction.
- **Implicit neural representation:** pioneered LTE (348+ citations) and a B-spline-based texture estimator, using Fourier and spline properties of the continuous domain to overcome the high-frequency limitation of coordinate-based neural representations, with downstream gains in super-resolution and OCR accuracy.
- **Diffusion models for generative imaging:** introduced training-free guidance methods (TAG, PAG) that amplify tangential/self-attention components during sampling to suppress hallucination, and a retrieval-augmented framework (RAG) for reference-based super-resolution.

HONORS AND AWARDS

- **2026:** ICBS (International Conference on Basic Science) Frontiers of Science Award.
- **2019:** IEEE Signal Processing Society (SPS) Best Paper Award, for “Deep Convolutional Neural Network for Inverse Problems in Imaging,” *IEEE Trans. on Image Processing*, 2017.
- **2015:** Grant, EPFL Fellows co-funded by Marie Skłodowska-Curie Actions (EU Horizon 2020); Samsung Humantech Paper Award – Silver Medal (Signal Processing).
- **2004:** The Presidential Science Scholarship, Ministry of Science and ICT, South Korea; Samsung Humantech Paper Award – Participation Prize.

ACADEMIC LEADERSHIP AND PROFESSIONAL SERVICE

- **Area Chair**, NeurIPS 2024-2026, ACCV 2026, ICLR 2027, ICML 2025-2026; **Senior Program Committee**, AAAI 2027.
- **Director and Executive/General Council Member**, Institute of Electronics and Engineers of Korea (IEIE), 2025–2026; Member, AI Signal & Image Processing Research Society (IEIE), 2022–2026.
- **Publication Chair**, ITC-CSCC 2025.
- **Program Committee**, IEIE Summer/Winter Annual Conference (2026), Image Processing Winter School (2024–2026), Joint Summer School on Image Understanding and Processing (2023–2025), and AI Signal

PEER-REVIEWED PUBLICATIONS

Research record: 37 peer-reviewed papers, 2021–2026 (plus 23 earlier, 2009–2020), including 20+ top-tier AI/vision conference papers (CVPR, ICCV, ECCV, ICML, ICLR, NeurIPS, WACV) since 2022 and a *Nature Communications* article (IF 14.7, 2024).

Representative venues: CVPR, ICCV, ECCV, ICML, ICLR, NeurIPS, WACV, MICCAI, ICRA (AI/vision); *Nature Communications*, *IEEE Trans. Image Processing*, *IEEE Trans. Medical Imaging*, *Pattern Recognition*, *Scientific Reports* (journals).

*Author's name underlined; [†] corresponding author; * equal contribution.*

- Byeonghun Lee, Hyunmin Cho, Sunghoon Im[†], and K. H. Jin[†], “ReAL: Reference-to-Image (R2I) Aware Latent Diffusion for Image Super-Resolution,” *European Conference on Computer Vision (ECCV)*, 2026, Accepted
- Hyunmin Cho, Jaejun Yoo, and K. H. Jin[†], “Recurrent Sinusoidal INRs for Efficient High-Fidelity Representation,” *European Conference on Computer Vision (ECCV)*, 2026, Accepted
- Soobin Park, Seohyeon Yoo, Jiwon Kim, SeonHwa Kim, K. H. Jin[†], and Eunju Cha[†], “SA-V2V: Training-free Subject-Aware Video-to-Video Personalization,” *European Conference on Computer Vision (ECCV)*, 2026, Accepted
- Y. Kim, K. H. Jin, Y. Han, and K. Ryu, “Improved Vascular Segmentation via Binary Flow Matching with Straight-Path Regularization,” *Medical Image Computing and Computer Assisted Interventions (MICCAI)*, 2026, Accepted
- Hyunmin Cho, Woo Kyoung Han, and K. H. Jin[†], “Balancing Fidelity and Diversity in Diffusion Models via Symmetric Attention Decomposition: Hopfield Perspective,” *International Conference on Machine Learning (ICML)*, 2026, Seoul
- Hyunmin Cho*, Donghoon Ahn*, Susung Hong*, Jee Eun Kim, Seungryong Kim[†], and K. H. Jin[†], “TAG: Tangential Amplifying Guidance for Hallucination-Resistant Sampling,” *International Conference on Machine Learning (ICML)*, 2026, Seoul
- Mingyu Choi, Woo Kyoung Han, Sunghoon Im[†], and K. H. Jin[†], “Linear Recurrent Unit with Semantic Modulation for Image Super-Resolution,” *IEEE/CVF Conference on Computer Vision and Pattern Recognition Findings (CVPR Findings)*, 2026, Denver
- Jiwon Kim, SeonHwa Kim, Soobin Park, Eunju Cha[†], and K. H. Jin[†], “Semantic Alignment for Pose-Invariant Identity Preserving Diffusion,” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026, Denver
- D. Ahn*, J. Kang*, S. Lee, J. Min, W. Jang, M. Kim, H. Cho, S. Paul, S. Kim, E. Cha[†], K. H. Jin[†], and Seungryong Kim[†], “A Noise is Worth Diffusion Guidance,” *International Conference on Learning Representations (ICLR)*, 2026
- Jung In Jang and K. H. Jin[†], “GRAPE (Gaussian Rendering for Accelerated Pixel Enhancement) Brings Fast and Lightweight Arbitrary Super-Resolution,” *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2026, Tucson
- Byeonghun Lee*, Hyunmin Cho*, Honggyu Choi, Soo Min Kang, Iljun Ahn, and K. H. Jin[†], “Reference-based Super-Resolution via Image-based Retrieval-Augmented Generation Diffusion,” *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025, Hawaii
- Woo Kyoung Han*, Yongjun Lee*, Byeonghun Lee, Sang Hyun Park, Sunghoon Im[†], and K. H. Jin[†], “JPEG Processing Neural Operator for Backward-Compatible Coding,” *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025, Hawaii
- Jiwon Kim, Pureum Kim, SeonHwa Kim, Soobin Park, Eunju Cha[†], and K. H. Jin[†], “Dual Recursive Feedback on Generation and Appearance Latents for Pose-Robust Text-to-Image Diffusion,” *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025, Hawaii
- Sejin Park, Sangmin Lee, K. H. Jin, and Seung-Won Jung, “IM-LUT: Interpolation Mixing Look-Up Tables for Image Super-Resolution,” *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025, Hawaii
- Woo Kyoung Han, Byeonghun Lee, Hyunmin Cho, Sunghoon Im[†], and K. H. Jin[†], “Towards Lossless Implicit Neural Representation via Bit Plane Decomposition,” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025

- SeonHwa Kim, Jiwon Kim, Soobin Park, Donghoon Ahn, Jiwon Kang, Seungryong Kim[†], K. H. Jin[†], and Eunju Cha[†], “Identity-preserving Distillation Sampling by Fixed-Point Iterator,” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025
- Eunjin Kim, Hyeonjin Kim, K. H. Jin, and Jaejun Yoo, “BF-STVSR: B-Splines and Fourier – Best Friends for High Fidelity Spatial-Temporal Video Super-Resolution,” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025
- EungGu Kang, Byeonghun Lee, Sunghoon Im[†], and K. H. Jin[†], “BurstM: Deep Burst Multi-scale SR using Fourier Space with Optical Flow,” *European Conference on Computer Vision (ECCV)*, 2024, Milano
- Donghoon Ahn*, Hyungwon Cho*, Jaewon Min, Wooseok Jang, Jungwoo Kim, SeonHwa Kim, Hyun Hee Park, K. H. Jin[†], and Seungryong Kim[†], “Self-Rectifying Diffusion Sampling with Perturbed-Attention Guidance,” *European Conference on Computer Vision (ECCV)*, 2024, Milano
- Woo Kyoung Han, Sunghoon Im, Jaedeok Kim[†], and K. H. Jin[†], “JDEC: JPEG Decoding via Enhanced Continuous Cosine Coefficients,” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024
- Minsu Kim, Giseop Kim, K. H. Jin, and Sunwook Choi, “BroadBEV: Collaborative LiDAR-camera Fusion for Broad-sighted Bird’s Eye View Map Construction,” *IEEE International Conference on Robotics and Automation (ICRA)*, 2024, (Oral presentation)
- Nari Hong, Boil Kim, Jaewon Lee, Han Kyoung Choe, K. H. Jin[†], and Hongki Kang[†], “Machine Learning-based High-frequency Neuronal Spike Reconstruction from Low-frequency and Low-sampling-rate Recordings,” *Nature Communications* 15, 635, 2024
- Minsu Kim, Yongjun Lee, Woo Kyoung Han, and K. H. Jin[†], “Learning Residual Elastic Warps for Image Stitching under Dirichlet Boundary Condition,” *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024
- Minsu Kim, Jaewon Lee, Byeonghun Lee, Sunghoon Im, and K. H. Jin[†], “Implicit Neural Image Stitching with Enhanced and Blended Feature Reconstruction,” *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024
- Jae-Won Lee, Jong-Hyun Won, Seonggwang Jeon, Yujin Choo, Yubin Yeon, Jin-Seon Oh, Minsoo Kim, SeonHwa Kim, InSuk Joung, Cheongjae Jang, Sung Jong Lee, Tae Hyun Kim, K. H. Jin, Giltae Song, Eun-Sol Kim, Jejoong Yoo, Eunok Paek, Yung-Kyun Noh*, and Keehyoung Joo*, “DeepFold: Enhancing Protein Structure Prediction through Optimized Loss Functions, Improved Template Features, and Re-optimized Energy Function,” *Bioinformatics* 39, no. 12, 2023
- Myeongkyun Kang, Philip Chikontwe, Soopil Kim, K. H. Jin, Ehsan Adeli, Kilian M. Pohl, and Sang Hyun Park, “One-shot Federated Learning on Medical Data using Knowledge Distillation with Image Synthesis and Client Model Adaptation,” *Medical Image Computing and Computer Assisted Interventions (MICCAI)*, 2023
- Woo Kyoung Han, Byeonghun Lee, Sang Hyun Park, and K. H. Jin[†], “ABCD: Arbitrary Bitwise Coefficient for De-quantization,” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023
- Byeonghyun Pak, Jaewon Lee, and K. H. Jin[†], “B-spline Texture Coefficients Estimator for Screen Content Image Super-Resolution,” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, **Highlight** (top 10% of accepted papers, top 2.5% of submissions)
- Jaewon Lee, Kwang Pyo Choi, and K. H. Jin[†], “Learning Local Implicit Fourier Representation for Image Warping,” *European Conference on Computer Vision (ECCV)*, 2022
- Jaewon Lee and K. H. Jin[†], “Local Texture Estimator for Implicit Representation Function,” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022
- Jaejun Yoo, K. H. Jin, H. Gupta, J. Yerly, M. Stuber, and M. Unser, “Time-dependent Deep Image Prior for Dynamic MRI,” *IEEE Trans. on Medical Imaging* 40(12): 3337–3348, 2021
- K. H. Jin, “Deep Block Transform for Autoencoders,” *IEEE Signal Processing Letters*, vol. 28, pp. 1016–1019, 2021
- **Before 2021, 23 additional peer-reviewed papers were published (early-career work on compressed sensing, annihilating-filter Hankel-matrix reconstruction, and terahertz imaging with J. C. Ye and D. Yee); full list available on request.**